

EE6221 Robotics and Intelligent Sensors

Beginner Notes: Manipulator Kinematics

Living course note

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Scope and sources. This self-contained note develops the supplied manipulator material from first vocabulary through direct and inverse kinematics, elementary trajectory planning, Jacobians, and singularities. It uses `resources/lecture-01-manipulator-kinematics.pdf`, pp. 1–156, `resources/course-outline-part-01.pdf`, and the Week 1/2 transcripts as supporting teaching context. Newly archived public slides, assignments, and examinations are recorded in `STATUS.md` but are not treated as covered here until independently reviewed.

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1 What is a robot manipulator?

For this course, a robot is an industrial, reprogrammable, multifunctional manipulator. A manipulator is a chain of rigid *links* joined by actuated *joints*; the final link carries an *end effector*, such as a gripper, hand, or tool. Moving the joints changes the end effector’s *pose*: its position and orientation.

The two fundamental joint types are:

- **Revolute (R):** the link rotates about an axis; its joint coordinate is an angle.
- **Prismatic (P):** the link slides along an axis; its joint coordinate is a displacement.

The number of independently controlled joint coordinates is the robot’s degrees of freedom (DoF). The first few joints often position the wrist, while wrist joints orient the tool. A robot can be useful with fewer than six DoF, but then it may be unable to realise arbitrary position-and-orientation requests.

2 Why and how robots are used

Robots are particularly useful for repetitive, hazardous, precise, or reconfigurable tasks. In contrast to a hard-automation machine designed for one fixed task, a reprogrammable robot can change task by changing its program and tooling. This flexibility has a cost: it may be less economical than dedicated equipment at very high production volumes, while manual work can be preferable at very low volumes.

Common manipulator structures are named from the major joint types: Cartesian PPP, cylindrical RPP, spherical RRP, SCARA RRP, and articulated RRR. Their structure affects the workspace: the set of tool poses that can be reached. The workspace is not simply a ball around the base. Joint limits, link dimensions, collisions, and orientation requirements restrict it.

Useful specifications include payload (allowable load), reach, stroke, speed, repeatability, accuracy, and resolution. *Accuracy* is closeness to the commanded pose; *repeatability* is how consistently the robot returns to a pose. A robot may be repeatable but inaccurate if a systematic calibration error shifts every placement.

Source: *resources/lecture-01-manipulator-kinematics.pdf*, pp. 1–22; *resources/week-01-transcript.txt*.

3 Why coordinate frames are essential

Saying “the tool is at $(1, 2, 3)$ ” is incomplete: those numbers only have meaning relative to a coordinate frame. We write $\{A\}$ for a base or reference frame and $\{B\}$ for a frame attached to a robot link. A physical vector is the same geometric object, but its components change when expressed in another frame.

If ${}^A R_B$ describes frame $\{B\}$ relative to frame $\{A\}$, then

$${}^A v = {}^A R_B {}^B v.$$

The columns of ${}^A R_B$ are the x_B , y_B , and z_B axes written in frame $\{A\}$. This gives an immediate interpretation: a rotation matrix tells an observer in $\{A\}$ which way the axes attached to $\{B\}$ point.

4 Rotations and homogeneous transformations

A valid three-dimensional rotation matrix satisfies

$$R^T R = I, \quad R^{-1} = R^T, \quad \det R = +1.$$

For example, a right-handed rotation about the z axis is

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Matrix order matters. Rotations about fixed axes premultiply an existing orientation; rotations about moving (body) axes postmultiply. Always state the frame and physical order instead of relying on a memorised product.

Rotation alone cannot account for different origins. If the origin of $\{B\}$ is at ${}^A p_B$, a point P satisfies

$${}^A p_P = {}^A R_B {}^B p_P + {}^A p_B.$$

Homogeneous coordinates combine both operations:

$${}^A T_B = \begin{bmatrix} {}^A R_B & {}^A p_B \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} {}^A p_P \\ 1 \end{bmatrix} = {}^A T_B \begin{bmatrix} {}^B p_P \\ 1 \end{bmatrix}.$$

The inverse is

$$({}^A T_B)^{-1} = \begin{bmatrix} ({}^A R_B)^T & -({}^A R_B)^T p_B \\ 0 & 1 \end{bmatrix}.$$

The translated term must be rotated when inverting; simply negating it is a common mistake.

Source: `resources/lecture-01-manipulator-kinematics.pdf`, pp. 23–52; `resources/week-01-transcript.txt`.

5 Direct kinematics and D–H construction

Direct kinematics asks: if the joint coordinates $q_1, \dots, q_n$ are known, where is the tool? The answer is the product

$${}^0 T_n(q) = {}^0 T_1(q_1) {}^1 T_2(q_2) \cdots {}^{n-1} T_n(q_n).$$

The superscripts and subscripts behave like frame labels: adjacent labels cancel in a valid chain. If a product does not follow a connected route from the source frame to the destination frame, its physical meaning is suspect.

The standard Denavit–Hartenberg construction represents each adjacent-link transform by four operations:

$${}^{i-1} T_i = R_z(\theta_i) T_z(d_i) T_x(a_i) R_x(\alpha_i).$$

Here θ_i rotates about z_{i-1} , d_i translates along z_{i-1} , a_i translates along x_i , and α_i rotates about x_i . For a revolute joint, θ_i is the variable; for a prismatic joint, d_i is the variable. The other quantities describe fixed link geometry.

Use this construction order:

1. Draw every joint axis and assign it as a z axis.
2. Choose each x_i along the common normal from z_{i-1} to z_i ; intersecting axes give $a_i = 0$.
3. Complete a right-handed frame with $y_i = z_i \times x_i$.
4. Fill the D–H table before multiplying matrices.
5. Check the final rotation block is orthonormal and the translation has length units.

Where each frame’s origin actually lands. The two translation parameters do not just appear in the matrix — they *move the origin*, and knowing where each origin ends up is what makes the table readable. Reading ${}^{k-1} T_k = \text{Rot}(z, \theta_k) \text{Trans}(z, d_k) \text{Trans}(x, a_k) \text{Rot}(x, \alpha_k)$ left to right: d_k slides the origin along z_{k-1} , then a_k slides it along the new x_k . So the translation column of ${}^{k-1} T_k$ is $(a_k c_k, a_k s_k, d_k)$, and frame k ’s origin sits a_k away from frame $k - 1$ ’s whenever $a_k \neq 0$.

For the articulated (RRR) waist/shoulder/elbow arm, with rows $(\theta_1, d_1, 0, 90^\circ)$, $(\theta_2, 0, a_2, 0)$, $(\theta_3, 0, a_3, 0)$, the origins therefore land on the physical landmarks:

- frame 1 — the **shoulder**, a height d_1 up the column ($a_1 = 0$, so no sideways step);
- frame 2 — the **elbow**, a distance a_2 out along link 2;
- frame 3 — the **wrist**, a distance a_3 out along link 3.

This is exactly why a_2 and a_3 are the link lengths: x_k runs along the link, so the step along x_k is the link.

Which z axes are joint axes — and why the last one is not. The rule is z_{k-1} is the axis of joint k , for $k = 1, \dots, n$. Count carefully: an n -joint arm has its joints described by $z_0, \dots, z_{n-1}$, so z_n **is not a joint axis** — there is no joint at the far end of the last link. For the three-joint articulated arm, z_0 is the waist axis, z_1 the shoulder axis, z_2 the elbow axis, and z_3 is the chain's free end.

Because nothing downstream constrains z_3 , the last row's α_n and d_n are yours to choose. Taking $\alpha_3 = 0$ gives $z_3 \parallel z_2$, which is the usual default simply because it keeps the table and the matrix simplest; $\alpha_3 = \pm 90^\circ$ points z_3 elsewhere and is equally correct, with the arm matrix changing to match.

What is *not* a free choice is that the arm reaches along x_3 : θ_3 swings x onto link 3 and $\text{Trans}(x, a_3)$ then walks along it, which is exactly why the tool length lives in the a column.

Common pitfall: the approach vector is a *tool-frame* idea. It is tempting to expect the last frame's z to point along the tool, because the convention $R = [r_1 \ r_2 \ r_3] = [\text{normal, sliding, approach}]$ puts the approach vector third, i.e. along z . That convention describes a *tool frame*, not a D–H link frame, so read the columns of 0R_n as $[x_n \ y_n \ z_n]$ unless you have deliberately built a tool frame.

Wanting z along the tool is reasonable — a pick-and-place arm matrix needs exactly that — but α_n cannot deliver it: $\text{Rot}(x, \alpha_n)$ holds x_n fixed and only spins y_n, z_n about it, so z_n can be z_{n-1} , or the perpendicular in the plane, but never x_n itself. That is why a gripper frame is an appended **fixed** rotation after ${}^{n-1}T_n$: an extra frame, not a joint, with no row in the D–H table.

Common pitfall. $a_k = 0$ does not mean frames $k-1$ and k coincide. It means frame k 's origin stays *on the z_{k-1} axis* — there is no common normal to step along, because the two joint axes intersect. The frames can still be separated by d_k along that axis, and they are still different frames with different orientations. In the arm above, $a_1 = 0$ puts the shoulder origin on the column axis; frames 1 and 2 remain a full link apart.

One row per joint — and a rigid tool is not a joint. The table has exactly as many rows as the arm has *joints*. It is tempting to give a row to every offset you can see in the drawing, but a fixed tool of length L_t bolted to the last link contributes no degree of freedom, so it must never occupy a joint row. Handle it in one of two ways, and say which you are doing:

- post-multiply the arm matrix by a constant $\text{Trans}(z, L_t)$, giving ${}^0T_{\text{tool}} = {}^0T_n \text{Trans}(z, L_t)$; or
 - add an explicitly labelled *fixed* frame after the last joint row, with no starred variable in it.
- If a tool row displaces a real joint, the chain silently loses a degree of freedom: the forward equations can no longer reach the poses the arm physically reaches, and the inverse solution will ask for a joint variable that the table never defined.

The thirty-second pre-flight check. Before you multiply anything, count. Mark each joint variable in the table with a star (θ_k^* or d_k^*), then count the stars and count the joints in the figure. The two numbers must agree. A cylindrical arm drawn with one revolute and two prismatic joints needs three starred variables — one θ and two d 's. If you have two stars and a constant L_t sitting in a row, a joint has gone missing. This check costs seconds and catches the single most expensive error in the question, because every later part inherits it.

A second consistency check is just as cheap: the α_k in your table and the $R_x(\alpha_k)$ you write in ${}^{k-1}T_k$ must carry *the same sign*. Writing $\alpha_2 = +90^\circ$ in the table and $R_x(-90^\circ)$ in the transform is worse than either error alone, because a marker cannot tell which one you meant and cannot award the benefit of the doubt.

For a planar two-revolute-link arm,

$$x = a_1 \cos q_1 + a_2 \cos(q_1 + q_2), \quad y = a_1 \sin q_1 + a_2 \sin(q_1 + q_2).$$

The second link uses the accumulated angle $q_1 + q_2$ because its orientation is measured after both joint rotations.

Source: *resources/lecture-01-manipulator-kinematics.pdf*, pp. 53–86..

6 Inverse kinematics: pose to joint coordinates

Inverse kinematics reverses the question: given a desired tool transform ${}^0T_n^*$, find q such that ${}^0T_n(q) = {}^0T_n^*$. This is not ordinary matrix inversion because q appears inside trigonometric functions and the mapping can be many-to-one.

Before solving equations, check three forms of feasibility:

- the desired position lies in the work envelope;
- the robot has enough orientation freedom for the desired tool orientation;
- each candidate respects joint limits and practical collision constraints.

An IK problem can have no solution, one solution, or multiple branches such as elbow-up and elbow-down. Use $\text{atan2}(y, x)$ when reconstructing an angle: unlike $\arctan(y/x)$, it preserves quadrant information and remains meaningful when $x = 0$.

Finish the answer: feasibility and how many solutions. Solving for the joint variables is only two-thirds of an inverse-kinematics answer. A complete solution also states the *range* over which it is valid and *how many* solutions there are. For a cylindrical arm with tool-tip target (x, y, z) and a radial slide d_3 behind a tool of length L_t ,

$$d_3 = \sqrt{x^2 + y^2} - L_t, \quad d_2 = z, \quad \theta_1 = \text{atan2}(\cdot, \cdot),$$

the answer is unfinished until you add: $d_3 \geq 0$, i.e. $\sqrt{x^2 + y^2} \geq L_t$ — points closer to the column than the tool is long are unreachable — and z must lie inside the column's stroke.

How many branches should you expect? Count the sub-chains, not the joints. Elbow-up/elbow-down pairs come from a *two-revolute (2R) sub-chain*, where two link lengths and one target distance form a triangle that can close in two mirror-image ways. A cylindrical arm has one revolute joint and no 2R sub-chain, so its solution is **unique**. Reporting a $\pm$ branch for d_3 imports a pattern from a different geometry: $d_3 + L_t = \sqrt{x^2 + y^2}$ is a length, so only the positive root is physical.

Analytic IK uses robot geometry to isolate variables and enumerate branches. Numerical IK instead iteratively reduces pose error, commonly through a Jacobian-based update. Numerical methods generalise better but need an initial guess, a convergence test, joint-limit handling, and a strategy near singularities.

Source: *resources/lecture-01-manipulator-kinematics.pdf*, pp. 87–117..

The tool-configuration vector: six numbers instead of twelve. Writing a tool pose as (p, R) carries twelve numbers to express six degrees of freedom, and the rotation matrix is awkward to use as the *input* to an inverse-kinematics problem. The handout therefore compresses orientation into the *approach vector* r_3 — the third column of R , the direction the tool advances — which fixes the tool's direction but says nothing about how far it has rolled about that direction. Since r_3 is a unit vector, its *length* is free, so the roll can be hidden in the magnitude:

$$w = \begin{bmatrix} p \\ e^{q_n/\pi} r_3 \end{bmatrix} \in \mathbb{R}^6,$$

where q_n is the **tool roll angle** — the rotation *about* the approach vector. The exponential is chosen because $e^{q_n/\pi} > 0$ always and is invertible, so the roll is recoverable: $q_n = \pi \ln \|w_{4:6}\|$.

Common pitfall. q_n is the roll *about* r_3 , not the first joint angle. Rotating the whole arm about its base column swings the approach vector to a new *direction*; it does not roll the tool about it, so it does not enter the exponent. An arm with no roll degree of freedom — a cylindrical RPP arm, for instance — has $q_n = 0$, hence $e^{q_n/\pi} = 1$ and $\|w_{4:6}\| = 1$. If your w comes out with $\|w_{4:6}\| \neq 1$ on an arm that cannot roll, the decode $q_n = \pi \ln\|w_{4:6}\|$ is claiming a roll the mechanism cannot perform.

Source: *resources/lecture-01-manipulator-kinematics.pdf*, pp. 91–95..

7 From a geometric path to a timed trajectory

A path specifies where the tool travels; a trajectory also specifies when it occupies each point. Writing $w(t) = \Gamma(s(t))$ separates the geometric curve Γ from the timing law $s(t)$. This distinction lets the same collision-free path be reused with slower or faster motion.

For one segment, a cubic polynomial

$$w(t) = at^3 + bt^2 + ct + d, \text{quad} 0 \leq t \leq T,$$

can satisfy two endpoint positions and two endpoint velocities. Componentwise, for $w(0) = w_0$, $\dot{w}(0) = v_0$, $w(T) = w_1$, and $\dot{w}(T) = v_1$,

$$\begin{aligned} d &= w_0, & c &= v_0, \\ a &= \frac{2(w_0 - w_1) + T(v_0 + v_1)}{T^3}, & b &= \frac{3(w_1 - w_0) - T(2v_0 + v_1)}{T^2}. \end{aligned}$$

The powers of T are a dimensional check: a has units of position/time³ and b position/time². Multi-segment motion should join with suitable velocity and acceleration continuity; matching positions alone can still create abrupt motion.

Building a grasp pose: which way do the jaws close? To write ${}^0T_{\text{pick}}$ for a parallel-jaw gripper you need two physical directions and then one cross product; guessing all three columns is how orthonormality gets lost.

1. **Approach** r_3 — the direction the tool *advances into* the part. Picking from above means descending, so $r_3 = -z_0 = (0, 0, -1)$, *not* $+z_0$. The approach vector points from the tool toward the workpiece.
2. **Sliding** r_2 — the direction the jaws *close*. This is perpendicular to the faces they touch. “Grasp the long sides” names the two faces being contacted; those faces are separated *across* the part, so the jaws travel along the *short* axis. If the long axis lies along x_0 , then $r_2 = (0, 1, 0)$. Choosing r_2 along the long axis instead would drive the jaws at the part’s short end faces.
3. **Normal** $r_1 = r_2 \times r_3$ — computed, never guessed. Then verify $r_1 \times r_2 = r_3$ and $\det R = +1$. A right-handed orthonormal set can still be the *wrong* frame for the task, so this check confirms validity, not correctness.
4. **Translation** — the centroid of the part *in its target location*. For a place pose that stacks part A on part B , the height is $z = z_B + \frac{1}{2}h_B + \frac{1}{2}h_A$: rise to B ’s top surface, then lift by A ’s own half-height. Both half-heights, or the part ends up buried in the one below it. Three terms, every time — and compute them from the dimensions in front of you, never from a number you remember from a similar problem.

The pick and the place are two different orientations. Run the recipe above *twice*, because r_2 depends on how the part is lying *at that moment*, and a place operation may well re-orient it:

- at the **pick**, use the part’s *current* long-axis direction;
- at the **place**, use its *target* long-axis direction — which is whatever the task demands, for example “aligned with the part underneath”.

If the two directions differ by 90° , so do the two gripper frames, and $R_{\text{place}} = R_z(\pm 90^\circ)R_{\text{pick}}$. If they are the same, R is unchanged and only the translation moves. Deciding which case you are in is the whole question — and both frames are perfectly valid rotations, so orthonormality and $\det R = +1$ will *not* catch the mistake of attaching the right matrix to the wrong pose.

The cheap habit: before writing either matrix, write the long-axis direction for that pose on its own line (“long $\parallel y_0$ ” then “long $\parallel x_0$ ”). The jaw direction is then forced — it is the perpendicular one — and the two poses cannot be interchanged by accident.

Source: *resources/lecture-01-manipulator-kinematics.pdf*, pp. 118–132; tool-frame column naming (normal/sliding/approach) pp. 57–58..

8 Jacobians, velocity control, and singularities

If the task coordinate is $x = w(q)$, differentiating with respect to time gives

$$\dot{x} = J(q)\dot{q}, \quad J(q) = \frac{\partial w}{\partial q}.$$

Column j of J is the instantaneous task-space motion produced by unit speed at joint j while all other joints are fixed. This is a local linear relationship at the current configuration, not an exact formula for a large finite displacement.

When an exact inverse does not exist, the Moore–Penrose pseudoinverse gives a least-squares or minimum-norm rate command:

$$\dot{q} = J^\dagger \dot{x}.$$

For a redundant robot, null-space motion $(I - J^\dagger J)z$ can change joint posture without changing the commanded first-order tool motion. This can help avoid limits or improve dexterity, but only if the secondary objective is chosen deliberately.

A singular configuration occurs when J loses rank. Then at least one task-space direction becomes unattainable instantaneously, and a pseudoinverse command may demand very large joint rates near the singularity. The smallest singular value of J is a useful warning: as it approaches zero, the mapping becomes ill-conditioned. In practice, slow down, re-plan, or use a damped least-squares inverse rather than trusting an unstable ordinary inverse.

Source: *resources/lecture-01-manipulator-kinematics.pdf*, pp. 133–156..

9 Full-course learning checklist

- Can I distinguish a link, joint, end effector, pose, and workspace?
- Can I explain why a reprogrammable robot differs from hard automation?
- Can I state the source and target frame for every rotation or transformation?
- Can I explain why the columns of a rotation matrix are axes expressed in another frame?
- Can I invert a homogeneous transform without forgetting to rotate the translation?
- Can I assign D–H frames, identify the variable parameter, and check the transform chain?
- Before solving IK, can I check reachability, orientation freedom, limits, and solution branches?
- Can I distinguish path geometry from trajectory timing and check cubic coefficient units?
- Can I explain $\dot{x} = J\dot{q}$ and what rank loss means physically?